

ROCKY MOUNTAIN HOUSE, AB, Canada

WMO: 719280

Lat: 52.4214N

Lon: 114.9122W

Elev: 988

StdP: 90.00

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.5	-28.5	-36.7	0.1	-31.9	-32.7	0.2	-28.0	9.2	-0.2	8.0	-2.4	1.1	120	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	26.7	16.4	24.9	15.5	23.3	14.8	17.8	24.4	16.7	23.2	15.6	21.6	3.5	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.3	12.2	21.1	14.1	11.3	19.5	13.1	10.6	18.2	54.3	24.5	50.6	23.4	47.2	21.8	22.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	6.8	5.9	-37.7	30.1	3.8	1.7	-40.4	31.3	-42.6	32.3	-44.7	33.3	-47.4	34.6		
(5)			-37.4	19.9	3.2	1.3	-39.7	20.8	-41.6	21.5	-43.4	22.2	-45.8	23.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.0	-11.1	-9.5	-4.9	2.5	8.3	12.2	15.1	13.9	9.3	3.1	-5.0	-10.7		
	DBStd	11.08	8.59	7.62	7.58	4.98	4.15	2.93	2.91	3.12	3.87	4.55	6.87	7.70		
	HDD10.0	3354	653	547	462	227	81	11	1	4	58	218	450	642		
	HDD18.3	5974	911	780	721	475	310	186	106	140	272	474	700	900		
	CDD10.0	431	0	0	0	2	29	76	159	126	37	2	0	0		
	CDD18.3	8	0	0	0	0	0	1	6	2	0	0	0	0		
	CDH23.3	356	0	0	0	1	26	35	152	112	31	1	0	0		
	CDH26.7	56	0	0	0	0	2	5	27	18	4	0	0	0		
(14) Wind	WSAvg	2.2	1.9	2.0	2.5	2.8	2.7	2.6	2.3	2.0	2.0	2.1	1.8	1.8		
(15) Precipitation	PrecAvg	541	22	19	24	31	63	107	90	73	45	24	21	19		
	PrecMax	647	58	39	40	88	108	192	207	137	98	55	35	43		
	PrecMin	405	6	6	4	8	23	22	17	21	4	5	6	5		
	PrecStd	79	11	9	9	20	25	42	36	29	27	14	10	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.4	10.1	13.0	21.3	26.2	26.6	29.2	28.7	26.6	21.3	14.0	9.1		
		MCWB	3.5	3.2	4.9	10.0	13.0	15.8	17.4	17.0	14.7	10.5	6.8	3.1		
	2%	DB	7.2	6.5	9.5	17.7	23.3	24.0	27.0	26.2	23.6	17.2	9.8	5.6		
		MCWB	1.9	1.2	3.2	8.3	11.8	14.5	17.5	16.7	13.9	9.1	4.1	0.9		
	5%	DB	4.1	3.7	7.2	14.7	20.7	22.0	24.9	24.2	21.1	14.1	6.6	2.8		
		MCWB	0.1	-0.5	1.9	6.5	10.8	13.5	16.5	15.9	13.0	7.4	2.0	-0.8		
	10%	DB	0.6	1.1	5.1	11.5	18.1	20.0	22.9	22.2	18.4	11.3	3.8	-0.1		
		MCWB	-1.7	-1.9	0.8	4.9	9.6	12.5	15.6	15.0	11.7	5.9	0.7	-2.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	3.5	5.7	10.2	14.1	17.1	19.8	18.9	16.1	11.7	7.1	3.3		
		MCDB	9.5	9.5	11.9	19.7	23.8	24.8	26.1	25.5	23.8	19.9	13.3	8.4		
	2%	WB	2.2	1.5	3.6	8.7	12.6	15.5	18.3	17.5	14.7	9.5	4.5	1.2		
		MCDB	7.1	6.0	8.7	16.8	21.4	21.8	24.9	24.2	22.2	16.2	9.1	5.3		
	5%	WB	0.4	-0.3	2.2	7.0	11.3	14.4	17.2	16.5	13.4	7.8	2.6	-0.6		
		MCDB	3.6	3.0	6.7	14.0	19.2	20.5	23.7	22.9	20.1	13.4	6.0	2.5		
	10%	WB	-1.7	-1.7	1.0	5.3	10.0	13.2	16.0	15.4	12.0	6.2	0.8	-2.4		
		MCDB	0.8	1.0	4.7	10.8	17.0	18.7	21.8	21.4	17.9	10.5	3.3	-0.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.3	13.5	12.0	12.5	13.7	12.5	13.1	13.7	14.0	13.0	11.5	11.9		
		MCDBR	15.7	16.4	15.3	19.0	19.4	17.1	17.0	18.0	20.4	18.6	15.2	14.4		
		MCWBR	11.2	11.6	9.7	10.7	9.7	8.7	8.7	9.5	11.5	10.9	9.9	10.4		
	5% WB	MCDBR	15.2	15.7	14.3	18.0	17.6	14.9	15.5	16.1	18.7	17.7	14.3	14.1		
		MCWBR	11.1	11.3	9.3	10.3	9.2	8.0	8.4	8.9	10.7	10.7	9.7	10.3		
(40) Clear-Sky Solar Irradiance	taub		0.256	0.261	0.296	0.331	0.344	0.353	0.350	0.360	0.318	0.283	0.255	0.242		
	taud		2.426	2.486	2.447	2.381	2.395	2.389	2.434	2.384	2.485	2.578	2.521	2.411		
	Ebn at Noon		774	884	906	906	907	899	895	864	870	846	779	727		
	Edh at Noon		54	69	88	107	111	113	106	106	84	63	49	46		
(44) All-Sky Solar Radiation	RadAvg		1.01	1.93	3.32	4.58	5.39	5.72	6.10	5.08	3.56	2.15	1.14	0.78		
	RadStd		0.06	0.12	0.20	0.36	0.38	0.43	0.38	0.33	0.39	0.22	0.09	0.05		

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page